Day-5

Assemble and Disassemble a PC practical.

:- Full hands on session :

-Assemble and Diassemble a Desktop PC.

-Document the process with labeled steps.

Q-1 :-List all the main components you find inside a desktop PC after opening the cabinet, and write a short description (1-2 lines) for each.

Ans-**Main Components Inside a Desktop PC Cabinet**

1. **Motherboard** – The main circuit board of the computer. It connects all the components and allows them to communicate with each other.

2. **CPU (Central Processing Unit)** – Known as the brain of the computer. It processes instructions and performs calculations.

3. **RAM (Random Access Memory)** – Temporarily stores the data and programs currently being used. More RAM helps the computer handle multiple tasks smoothly.

4. **Hard Disk Drive (HDD) / Solid State Drive (SSD)** – Stores the operating system, software, documents, photos, and other files. An SSD is generally faster than an HDD.

5. **Power Supply Unit (PSU)** – Converts electricity from the wall socket into the correct power needed by the computer components.

6. **CPU Cooler / Heat Sink** – Removes heat from the processor to prevent it from overheating. Some computers use a fan, while others use liquid cooling.

7. **Cabinet Fans** – Move air inside the cabinet to keep the components cool and prevent overheating.

8. **Graphics Card (GPU)** – Processes images, videos, and graphics and sends them to the monitor. Some computers have a separate graphics card, while others use built-in graphics.

9. **Optical Drive (CD/DVD Drive)** – Reads CDs and DVDs and, in some models, writes data to them. It is optional in modern desktop computers.

10. **Expansion Cards** – Additional cards fitted into the motherboard to provide extra features, such as better sound, network connections, or more USB ports.

11. **CMOS Battery** – A small, round battery on the motherboard. It helps maintain the computer's clock and BIOS settings when the power is switched off.

12. **BIOS/UEFI Chip** – Stores the basic firmware that helps the computer start up and checks the hardware before loading the operating system.

13. **SATA and Power Cables** – Connect storage drives and other components to the motherboard and power supply so they can work properly.

14. **Network Card (LAN/Wi-Fi)** – Allows the computer to connect to a network or the internet. It may be built into the motherboard or installed as a separate card.

15. **Sound Card / Audio Chip** – Handles sound input and output, allowing the computer to play audio through speakers or headphones and record sound through a microphone.

16. **Front Panel Connectors** – Connect the cabinet's power button, restart button, indicator lights, and front USB and audio ports to the motherboard.

17. **Cabinet (Computer Case)** – The outer enclosure that holds and protects all the internal components and helps provide proper airflow.

Note: Not every desktop PC contains all these components. For example, a separate graphics card, sound card, optical drive, or network card may not be present in every computer.

Q-2 :-Take clear photos or draw labeled diagrams of each step as you disassemble a desktop PC, and organize them in a document with captions explaining what you did at each step.

Hint: Include steps like unplugging power, removing side panel, disconnecting cables, and removing RAM, HDD/SSD, and motherboard.

Ans-**Desktop PC Disassembly – Step-by-Step**

- To disassemble a desktop computer safely and identify its main components.

Tools required: Screwdriver, small container for screws.

Step 1: Unplug the Power Cable



Switch off the computer and unplug the power cable from the wall socket. Disconnect all other cables.

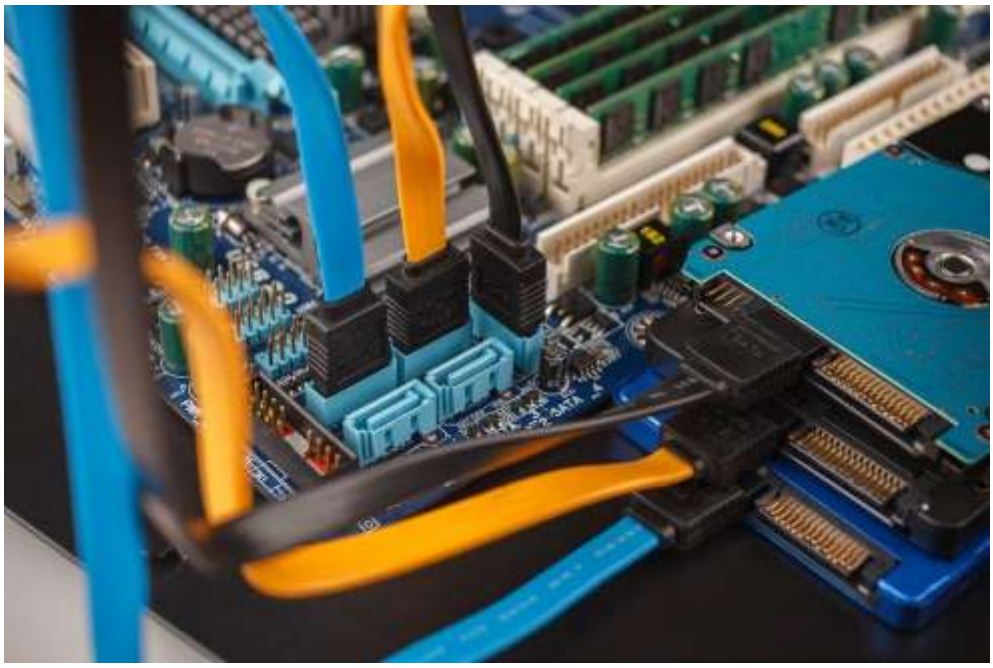
Safety note: The computer is disconnected from the power supply for safety.

Step 2: Remove the Side Panel



Unscrew the side panel of the cabinet and carefully slide it off.
Note: The side panel is removed to access the internal components.

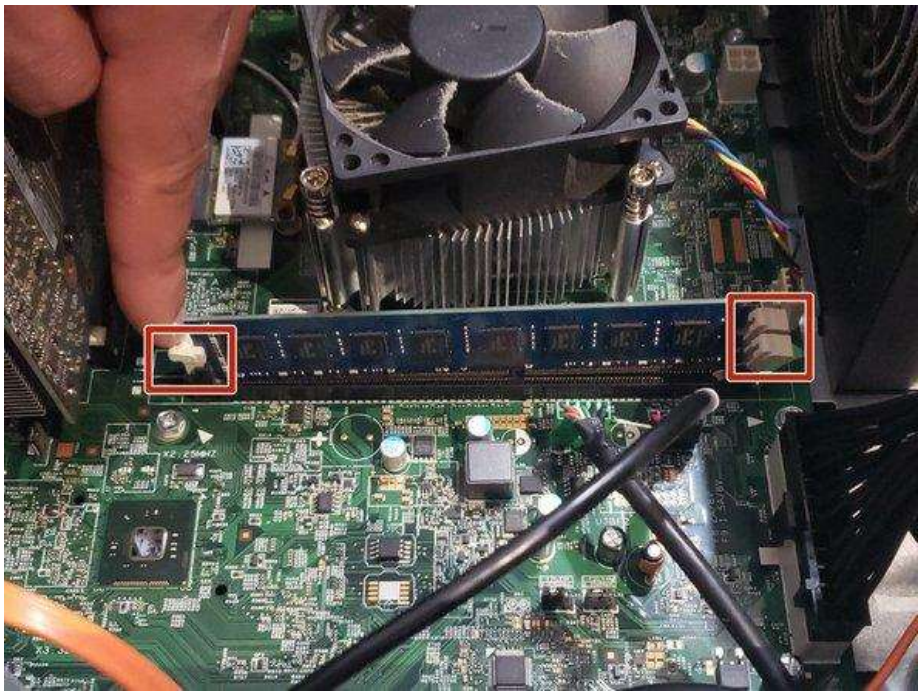
Step 3: Disconnect the Cables



Carefully disconnect the power and data cables from the components. Take a photo before removing them.

Note: The cables are disconnected carefully before removing the components.

Step 4: Remove the RAM



Open the clips at the ends of the RAM slot and gently remove the RAM stick.
Note: The RAM is removed from its slot on the motherboard.

tep 5: Remove the HDD/SSD



Disconnect its cables, remove the screws or mounting clips, and carefully take out the storage drive.
Note: The storage drive is removed from the cabinet.

Step 6: Remove the Power Supply



After disconnecting all its cables, unscrew the power supply unit and carefully remove it. Note: The power supply is removed from the cabinet.

Step 7: Remove the Motherboard



Make sure all cables and attached components are disconnected. Remove the screws and carefully lift out the motherboard.
Note: The motherboard is removed carefully from the cabinet.

Step 8: Arrange All Components



Place the removed parts neatly on a safe surface and identify each component.
Note: All the main components are separated and ready for identification.

Safety Precautions

-Always unplug the computer before opening the cabinet.
-Touch a grounded metal surface before handling components to reduce static electricity.
-Handle the RAM, motherboard, and other parts gently by their edges.
-Never open the power supply unit itself, as it can contain dangerous electrical charges.

Conclusion: The desktop PC was disassembled step by step, and its main components were identified.